

Gunter & Scott 1990 — Inventory of Definitions and Theorems

Source: **C. A. Gunter and D. S. Scott**, “**Semantic Domains**,” *Handbook of Theoretical Computer Science* Vol. B, North-Holland, 1990, pp. 633–674 ([../papers/Gunter Scott 1990.pdf](#)).

The work list for the Lean formalization: every definition and every one of the paper’s **30 numbered results** (Theorems / Lemmas / Proposition 1–30), in paper order, matched to its Lean equivalent.

Progress (as of r0009, 2026-0806)

#	Quantity	Done	Remaining	Of
1	Definitions to define	8	5	≈13
2	Numbered results complete	7 (Thm 1, Thm 3, Lem 4, Lem 5, Thm 6, Thm 7, Lem 8)	22	29
3	Unnumbered prose claims proved	11	—	—
4	Mathlib foundations reused	12	—	12
5	Theorems in the development	122 live (+6 commented out as unused)	—	—
6	sorry in the development	—	0	—

Row 2 counts only the paper’s 30 **numbered** results (Theorems / Lemmas / Proposition 1–30). Row 3 counts the claims the paper makes **in prose** rather than as numbered results; these are paper content too, and all eleven are formally verified:

#	Paper claim	Where	Lean
1	“the compact elements [of $P \rightarrow N$] are just the finite subsets of N ”	p. 9	<code>isCompactElement_iff_finite</code>
2	“ $P \rightarrow N$... is a domain”	p. 9	<code>instance : Domain (Set X)</code>
3	“ $D \rightarrow E$ is a ... cpo”	Thm 7 proof	<code>instance : CompletePartialOrder (ScottHom α β)</code>
4	“... bounded complete ... whenever E is”	Thm 7 proof	<code>instance : BoundedComplete (ScottHom α β)</code>
5	“step(s) ... is continuous”	Thm 7 proof	<code>scottContinuous_stepFun</code>
6	“... and compact in the ordering on $D \rightarrow E$ ”	Thm 7 proof	<code>isCompactElement_step</code>
7	“they form a basis for $D \rightarrow E$ ”	Thm 7 proof	<code>instance : IsAlgebraic (ScottHom α β)</code>
8	every compact function is a <i>finite</i> join of step functions	Thm 7 proof, implicit	<code>exists_finite_isLUB_of_isCo</code>
9	“an embedding is an injection”	§3.1	<code>IsEmbeddingProjectionPair.i</code>
10	“a projection is a surjection”	§3.1	<code>IsEmbeddingProjectionPair.s</code>
11	“it is easy to check that p_N ... is a finitary projection”	§3.1	<code>isFinitaryProjection_normal</code>

Six of those eight are the body of **Theorem 7**, which is now **complete** — all four conjuncts of its conclusion (cpo, bounded complete, algebraic, countably based) are formally verified, as `ScottHom.isBoundedCompleteDomain_scottHom`.

It is proved under **weaker hypotheses than the paper states**: bounded completeness of D is never used. D need only be a domain. The function space is a cpo for any preordered D , algebraic when D and E are, bounded complete because E is, and countably based because D and E are.

The remaining theorems are supporting API: the $\ll$ calculus, the `compactsBelow` machinery, the pointwise order and `suprema` on $D \rightarrow E$, and the step-function adjunction. The paper assumes or elides all of it.

Reference audit (r0020). Of the theorems in the development, 16 are never cited elsewhere. Nine of those are *terminal by design* — they are the paper’s own claims, so nothing should cite them (`injective_embedding`, `surjective_projection`, the `isNormalIn_sUnion*` family and `mono_right` for Lemma 4, `singleton_bot_isNormalIn` for $\{\perp\} \in P(C)$, `isLeast_kleeneFix_le`, `eq_kleeneOperator_op`). The other six were speculative API written for callers that never appeared; they are **commented out in place**, each with a note on why it exists and what is instructive about it, and the build is unchanged — which also confirms that the three `@[simp]` ones among them were never firing implicitly.

4 of ≈ 13 definitions are formally verified and building, in 6 modules, 985 lines, 0 sorry, 0 warnings:

#	Round	Module	Contents
1	r0003	<code>ScottDomains/WayBelow.lean</code>	<code>way-below</code> $\ll$ at $[Preorder\ \alpha]$, 7 theorems; $x \ll x \leftrightarrow IsCompactElement\ x$ holds by <code>Iff.rfl</code>
2	r0004	<code>ScottDomains/Domain.lean</code>	<code>IsAlgebraic</code> , <code>Domain</code> (with the paper’s countable-basis condition), <code>BoundedComplete</code> , 8 theorems, <code>Domain Prop</code>
3	r0005	<code>ScottDomains/Powerset.lean</code>	the paper’s $P\ N$ (p. 9): compacts of <code>Set X</code> are exactly the finite subsets, hence <code>Domain (Set N)</code> — the nondegenerate witness
4	r0006–r0007	<code>ScottDomains/ScottHom.lean</code>	the continuous function space $D \rightarrow E$: <code>ScottHom</code> , the pointwise order, <code>CompletePartialOrder</code> , and <code>BoundedComplete</code> when E is — Theorem 7’s first sentence in full
5	r0008	<code>ScottDomains/StepFunction.lean</code>	the single step function <code>step k e</code> : continuity (from k compact), the adjunction <code>step k e ≤ f ↔ e ≤ f k</code> , and compactness in $D \rightarrow E$ (from e compact)
6	r0009	<code>ScottDomains/FunctionSpaceDomain.lean</code>	<code>IsAlgebraic</code> — the paper’s “they form a basis for $D \rightarrow E$ ”; the two halves of <code>IsAlgebraic</code> use disjoint hypotheses (directedness needs only E bounded complete; the <code>lub</code> needs only D, E algebraic)
7	r0010	<code>ScottDomains/CompactFunctionSpace.lean</code>	every compact function is a finite join of step functions — the finiteness half of the basis claim
8	r0011	<code>ScottDomains/FunctionSpaceCountable.lean</code>	<code>Countable</code> — $K(D, E)$ is countable, hence <code>Domain (ScottHom $\alpha\ \beta$)</code> — Theorem 7 complete
9	r0012	<code>ScottDomains/NormalSubposet.lean</code>	the normal-subposet relation $\triangleleft$ and Lemma 4 , all four parts
10	r0012–r0013	<code>ScottDomains/Projection.lean</code>	embedding–projection pairs, projections, the paper’s “an embedding is an injection, a projection is a surjection”; <code>im(p)</code> is a cpo and finitary projections

11 | r0014 | `ScottDomains/FinitaryProjection.lean` | **Lemma 5** — the compacts of `im(p)` are `im(p) ∩ K(D)` (needs only that p is a projection), and `im(p) ∩ K(D)  $\triangleleft$  K(D)` |

12 | r0015 | `ScottDomains/NormalProjection.lean` | $p_N(x) = \bigsqcup \{y \in N \mid y \sqsubseteq x\}$: continuous, a projection, and `im(p_N) ∩ K(D) = N` — half of **Theorem 6**’s correspondence, plus order preservation both ways |

13 | r0016 | ScottDomains/Theorem6.lean | **Theorem 6** — p_N is finitary, $p_{\{im(p) \cap K(D)\}} = p$, and the correspondence assembled |

14 | r0017 | ScottDomains/FixedPoint.lean | **Theorem 1** — $\bigsqcup_n f^n(\perp)$ is the least fixed point of a continuous f on a cpo. Not Mathlib reuse; see the §2 table |

15 | r0018 | ScottDomains/UniformFixedPoint.lean | **Theorem 3** — fix is the unique uniform fixed-point operator; $\uparrow a$ as a cpo |

16 | r0019 | ScottDomains/Product.lean | $D \times E$ as a cpo (the one construction needing no case split) and **Lemma 8 parts 1–3** |

§2 and §3.1 are now complete (§3.2’s effective presentations excepted).

17 | r0021 | ScottDomains/Currying.lean | **Lemma 8.4** — $currying, D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$, and with it **Lemma 8 complete** |

18 | r0022 | ScottDomains/EffectivePresentation.lean | §3.2’s **effective presentation** — the enumeration of the basis with its two decidability conditions. **§3 complete** |

§2 and §3 are now complete — the only §3 omission is the paper’s *computable function*, which needs an r.e.-predicate notion Mathlib does not supply.

Next: the smash product $D \otimes E$, sum $D + E$ and lift $D\perp$, then Lemma 9 (the strict analogues of Lemma 8) and Lemma 10 (closure of bounded completeness).

Work counts

- **Reuse from Mathlib — no work (11):** poset, directed, cpo, $\perp$, monotone, continuous, fixed-point operator, Schröder–Bernstein (2), algebraic lattice, product, λ -notation. (Theorem 1 was listed here in an earlier draft and has been removed: Mathlib’s `OrderHom.lfp` is Knaster–Tarski over a complete lattice, not Kleene’s $\bigsqcup_n f^n(\perp)$ over a cpo. It is proved in `ScottDomains/FixedPoint.lean`, so 29 numbered results needed proof, not 28.)
- **Generalize / adapt (4):** compact element (`IsCompactElement` — its *definition* is already stated at `[PartialOrder  $\alpha$ ]` and so applies to a dcpo verbatim; what is `CompleteLattice`-only is every lemma about it, so the dcpo API must be re-proved), sum, lift (`WithBot/Sum`), ideal completion (`Order.Ideal`).
- **Definitions to define — new (≈ 13 ; 4 done in r0003–r0004, 9 remaining):** way-below $\ll$ (**done** — `ScottDomains/WayBelow.lean`), algebraic cpo, **domain**, bounded-complete (**all three done** — `ScottDomains/Domain.lean`), embedding–projection pair, (finitary) projection, normal subposet, effective presentation, smash product, the three powerdomains (Hoare / Smyth / Plotkin), bifinite / Plotkin order, and D^∞ .
- **Theorems to prove (28):** 28 of the paper’s 30 numbered results — Theorems 3, 6, 7, 11, 12, 14, 16, 18, 21, 22, 25, 26, 27, 29; Lemmas 4, 5, 8–10, 13, 17, 19, 20, 23, 24, 28, 30; Proposition 15. Only Theorems 1 & 2 come free from Mathlib.

Bottom line: ≈ 13 definitions to define + 28 results to prove, on top of 12 reused Mathlib foundations. After r0004: 9 definitions + 28 results remain.

Lean column legend: ✓ reuse Mathlib (name given) · ~ partial (Mathlib has a related or lattice-only version — generalize) · ✕ define / prove (absent from Mathlib v4.32.2, confirmed by grep).

Statements are paraphrased/de-garbled from the PDF (1990 Type-3 fonts render $\rightarrow$ as $!$, drop *fi* ligatures, mangle math), so read them as a guide to *what* each result is. Notation: $\sqsubseteq$ order, $\bigsqcup$ directed sup, $\perp$ bottom, $\ll$ way-below, $K(D)$ compacts, $Fp(D)$ finitary projections.

§2 Recursive definitions of functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	poset (partially ordered set)	✓ <code>PartialOrder</code>
2.1	Def	—	directed subset M : every finite $u \subseteq M$ has an upper bound in M	✓ <code>DirectedOn</code> / <code>IsDirected</code>
2.1	Def	—	cpo : poset in which every directed M has a lub $\bigsqcup M$	✓ <code>CompletePartialOrder</code> (chains: <code>OmegaCompletePartialOrder</code>)
2.1	Def	—	bottom $\perp$ (least element)	✓ <code>OrderBot</code>

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
2.1	Def	—	monotone function	✓ Monotone / OrderHom
2.1	Def	—	continuous : $\text{monotone}, f(\bigsqcup M) = \bigsqcup f(M)$ for directed M	✓ ScottContinuous
2.1	Thm	1	Fixed-Point Theorem : $f : D \rightarrow D$ continuous $\implies$ least fixed point $\bigsqcup_n f^n(\perp)$	✓ proved (r0017) — ScottDomains.theorem1. Not Mathlib reuse: OrderHom.lfp is Knaster–Tarski over a <i>complete lattice</i> with only monotonicity, a different theorem
2.2	Thm	2	Schröder–Bernstein for sets	✓ Function.schroeder_bernstein (SetTheory/Cardinal/SchroederBernstein.lean — the name in an earlier draft of this row, Function.Embedding.schroederBernstein, does not exist
2.3	Def	—	fixed-point operator (uniform)	✓ OrderHom.lfp / LawfulFix
2.3	Thm	3	The standard operator is the unique uniform fixed-point operator	✓ proved (r0018) — ScottDomains.theorem3; a fixed point operator is formalized as a family over every CompletePartialOrder in a universe

§3 Effectively presented domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	compact element $x: x \sqsubseteq \bigsqcup M (\text{dir.}) \implies x \sqsubseteq y \text{ some } y \in M; K(D)$	~ IsCompactElement — def is [PartialOrder], so reusable on a dcpo; its lemmas are all CompleteLattice-only
3.1	Def	—	way-below $\ll$ (approximation relation behind compactness)	✓ ScottDomains.WayBelow (r0003) — $x \ll x \leftrightarrow \text{IsCompactElement } x$ by Iff.rfl
3.1	Def	—	algebraic cpo: $x = \bigsqcup \{x' \in K(D) : x' \sqsubseteq x\}$ (directed)	✓ ScottDomains.IsAlgebraic (r0004)
3.1	Def	—	domain = algebraic cpo whose basis $K(D)$ is countable (the paper’s definition, p. 9 — the countability condition was missing from an earlier draft of this row)	✓ ScottDomains.Domain (r0004)
3.1	Def	—	bounded complete : $\perp +$ every bounded subset has a sup — a <i>separate</i> predicate; the paper composes them as “bounded complete domain” (Thm 7, Lem 10, Lem 13, Thm 14), which is the literature’s <i>Scott domain</i>	✓ ScottDomains.BoundedComplete (r0004); the compound is [Domain α] [BoundedComplete α]
3.1	Def	—	(countably based) algebraic lattice	✓ IsCompactlyGenerated (+CompleteLattice)
3.1	Def	—	embedding–projection pair (g, f)	✓ ScottHom.IsEmbeddingProjectionPair (r0012)
3.1	Def	—	projection; finitary projection p : $p \circ p = p \sqsubseteq \text{id}$, $\text{im}(p)$ a domain	✓ ScottHom.IsProjection (r0012), ScottHom.IsFinitaryProjection (r0013) — $\text{im}(p)$ carries a CompletePartialOrder via IsProjection.rangeCompletePartialOrder

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
3.1	Def	—	normal subposet / substructure	✓ <code>ScottDomains.IsNormalIn</code> , notation $\triangleleft$ (r0012)
3.1	Lem	4	$\langle P(C), \triangleleft \rangle$ of substructures is a cpo with $\{\perp\}$ least	✓ proved (r0012), all four parts
3.1	Lem	5	p finitary projection $\implies$ compacts of $\text{im}(p)$ are $\text{im}(p) \cap K(D)$, and $\text{im}(p) \cap K(D) \triangleleft K(D)$	✓ proved (r0014) — <code>IsProjection.isCompactElement_iff</code> (needs only <i>projection</i>) and <code>IsFinitaryProjection.isNormalIn_compacts</code>
3.1	Thm	6	Isomorphism: normal substructures $\cong \text{Fp}(D)$ (finitary projections)	✓ proved (r0015–r0016) — <code>ScottDomains.theorem6</code>
3.1	Thm	7	D, E bounded-complete domains $\implies D \rightarrow E$ bounded-complete domain	✓ proved (r0006–r0011) — <code>ScottHom.isBoundedCompleteDomain_scottHom</code> ; D bounded complete is not needed
3.2	Def	—	effective presentation $d : \mathbb{N} \rightarrow K(D)$; effectively presented domain	✓ <code>ScottDomains.EffectivePresentation</code> (r0022). The paper’s computable function is <i>not</i> formalized: it needs “recursively enumerable”, and Mathlib v4.32.2 has no <code>RePred</code> or equivalent (grep finds none) — building it is recursion theory, not domain theory

§4 Operators and functions

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
4.1	Def	—	product $D \times E$	✓ Prod order from Mathlib; the cpo instance is <code>ScottDomains.instCompletePartialOrderProd</code> (r0019) — Mathlib has <code>Prod.supSet</code> and <code>isLUB_prod</code> but no cpo instance
4.2	Def	—	Church’s λ-notation (continuous abstraction)	✓ <code>OrderHom / ωCPO ContinuousHom</code>
4.3	Def	—	smash product $D \otimes E$	✗ define
4.4	Def	—	sum $D + E$; lift $D\perp$	~ <code>Sum / WithBot, Part</code> (partial)
4.x	Lem	8	$D \times E \cong E \times D$; $(D \times E) \times F \cong D \times (E \times F)$; $D \rightarrow (E \times F) \cong (D \rightarrow E) \times (D \rightarrow F)$; $D \rightarrow (E \rightarrow F) \cong (D \times E) \rightarrow F$	✓ proved — <code>prodComm</code> , <code>prodAssoc</code> , <code>scottHomProd</code> (r0019); <code>scottHomCurry</code> (r0021)
4.x	Lem	9	Product/function-space iso laws over D, E, F	✗ prove
4.5	Lem	10	D, E bounded complete $\implies \rightarrow, \times, \otimes, +, ()\perp$ bounded complete	✗ prove
4.5	Thm	11	Ideal completion of a countable pre-order is a domain (all domains so arise)	~ <code>Order.Ideal</code> exists $\rightarrow$ prove
4.5	Thm	12	Initiality of a continuous algebra satisfying axioms T	✗ prove
4.5	Lem	13	D bounded complete $\implies$ powerdomains $D], D[$ bounded complete	✗ prove
4.5	Thm	14	Equivalent characterizations of an (algebraic/BC) domain	✗ prove

§5 Powerdomains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
5.1	Def	—	powerdomain (non-deterministic outcomes)	✗ define
5.2	Def	—	Hoare (lower), Smyth (upper), Plotkin (convex) powerdomains	✗ define
5.3	—	—	Universal & closure properties (see Lem 13, 28, 30)	✗ prove

§6 Bifinite (SFP) domains

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
6.1	Def	—	Plotkin order / bifinite (SFP) domain: bilimit of finite pointed posets	✗ define
6.1	Prop	15	Every bounded-complete domain is bifinite	✗ prove
6.1	Thm	16	D bifinite $\implies \text{Fp}(D)$ is an algebraic lattice	✗ prove
6.2	Lem	17	D, E bifinite $\implies \rightarrow, \times, \otimes, +, ()^\perp$ bifinite (incl. function space)	✗ prove
6.2	Thm	18	If D and $D \rightarrow D$ are domains, then D is bifinite	✗ prove
6.2	Lem	19	closure $r : D \rightarrow D$ ($r \circ r = r \sqcup \text{id}$) $\implies \text{im}(r)$ is a domain	✗ prove
6.2	Lem	20	D domain $\implies \text{Fc}(D)$ (finitary closures) is a cpo	✗ prove

§7 Recursive definitions of domains (universal domain, D_∞)

§	Kind	Ref	Concept / statement	In Lean / Mathlib?
7	Def	—	recursive domain equation; universal domain / D_∞ (inverse limit)	✗ define
7	Thm	21	F representable over cpo $U \implies$ a domain D with $D \cong F(D)$	✗ prove
7	Thm	22	Any countably-based algebraic lattice L : a closure $r : P(\mathbb{N}) \rightarrow L$	$\sim \text{IsCompactlyGenerated} \rightarrow$ prove
7	Lem	23	The function-space operator is representable over $P(\mathbb{N})$	✗ prove
7	Lem	24	U cpo; $\times$ and $\rightarrow$ representable over $U \implies$ (setup for universality)	✗ prove
7	Thm	25	U non-trivial domain representing $\times, \rightarrow \implies U$ universal	✗ prove
7	Thm	26	Any signature $(s_1, \dots, s_n)$: combinators $F_1, \dots, F_n$ solving the equations	✗ prove
7	Thm	27	Any bounded-complete D : a projection of the universal domain onto D	✗ prove
7	Lem	28	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ representable over U	✗ prove
7	Thm	29	D bifinite $\implies D^+$ bifinite; solving $D \cong D^+$	✗ prove
7	Lem	30	Operators $\rightarrow, \times, \otimes, +, ()^\perp, ()], ()[$ p-representable over universal bifinite V	✗ prove

Tally (matched): ✓ reuse Mathlib $\approx$ **12** (poset, directed, cpo, $\perp$, monotone, continuous, fixed-point Thm 1 & operator, Schröder–Bernstein, algebraic lattice, product, λ -notation) · $\sim$ partial $\approx$ **4** (compact element, sum/lift, ideal completion Thm 11, Thm 22) · $\times$ define / prove $\approx$ **44**, of which 4 are done ($\ll$ r0003; algebraic, domain, bounded-complete r0004) $\rightarrow$ **40 remaining** — the bifinite / powerdomain / D^∞ development and all 28 numbered results.

What is done, and what is next, is listed under **Progress** at the top of this file.